

Geometry

A Modern Introduction

Roman Steinberger

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Preface

Geometry is the study of the properties and relationships of figures in the plane and in space. Since the time of Euclid, it has served as one of the principal means by which students learn the nature of logical reasoning and mathematical proof. From a small collection of definitions, postulates, and previously established results, an extensive body of knowledge is developed through careful deduction.

The purpose of this book is to present the fundamental principles of Euclidean geometry in a clear, orderly, and systematic manner. Each topic is introduced with precise definitions and illustrated by examples and diagrams before being reinforced through exercises of varying difficulty. Throughout the text, emphasis is placed not only on obtaining correct answers but also on understanding the reasoning that justifies them.

Geometry is more than the study of shapes and measurements. It provides a framework for analyzing patterns, making logical arguments, and solving problems with precision. These skills are of lasting value in mathematics, the sciences, engineering, architecture, and many other disciplines.

The reader is encouraged to approach each new topic thoughtfully, to examine every diagram carefully, and to develop the habit of proving statements rather than accepting them by intuition alone. Mastery of geometry is achieved through patient study, regular practice, and an appreciation for the logical structure that unifies the subject.

It is hoped that this book will provide a firm foundation in geometry and foster an enduring appreciation for the elegance, rigor, and usefulness of mathematical reasoning.

Chapter 1

Essentials of Geometry

1.1 Points, Lines, and Planes	1
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Perimeter

The distance around a rectangle is called its **perimeter**.

Circumference

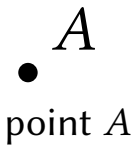
The distance around a circle is called its **circumference**.

1.1 Points, Lines, and Planes

Undefined Terms

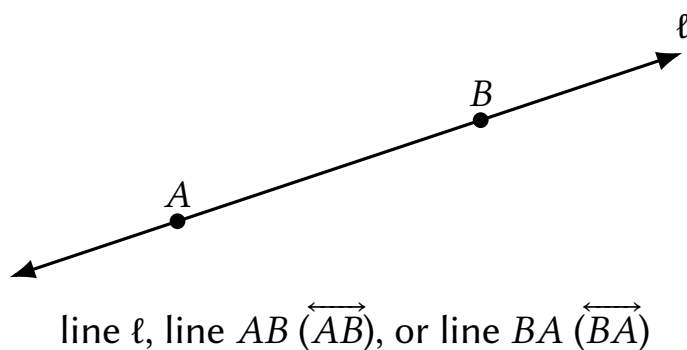
Undefined terms: In geometry, the words *point*, *line*, and *plane* are undefined terms. These words do not have formal definitions, but there is agreement about what they mean.

Point: A point has no dimension. It is represented by a dot.



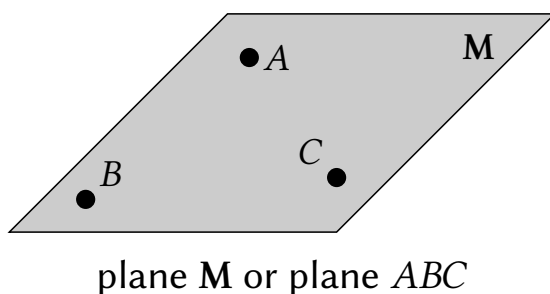
Line: A line has one dimension. It is represented by a line with two arrowheads, but it extends without end.

Through any two points, there is exactly one line. You can use any two points on a line to name it.



Plane: A plane has two dimensions. It is represented by a shape that looks like a floor or a wall, but it extends without end.

Through any three points not on the same line, there is exactly one plane. You can use three points that are not all on the same line to name a plane.

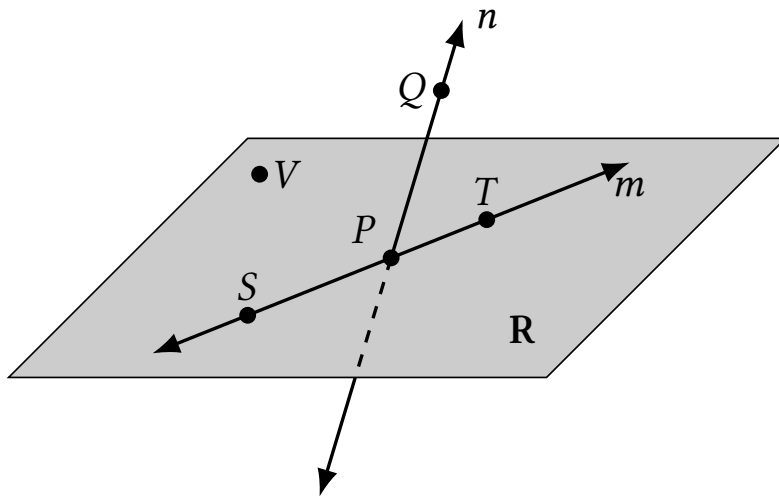


Collinear points: Collinear points are points that lie on the same line.

Coplanar points: Coplanar points are points that lie in the same plane.

Example 1

Name points, lines, and planes



- (a) Give two other names for $\overleftrightarrow{PQ}$
- (b) Give two other names for plane R.
- (c) Name three points that are collinear.
- (d) Name four points that are coplanar.
- (e) Give two other names for $\overleftrightarrow{ST}$.
- (f) Name a point that is not coplanar with points Q, S, and T.

Solution

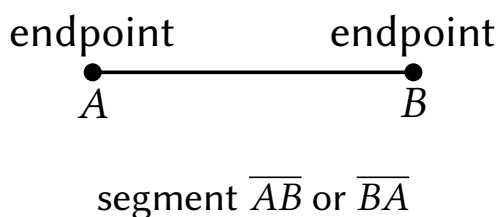
- (a) Line n , $\overleftrightarrow{QP}$.
- (b) Plane VST , plane VPT .
- (c) Points S , P , and T .
- (d) Points V , S , P , and T .
- (e) Line m , $\overleftrightarrow{PT}$.
- (f) Point V .

Defined Terms

Defined terms: In geometry, terms that can be described using known words such as *point* or *line* are called defined terms.

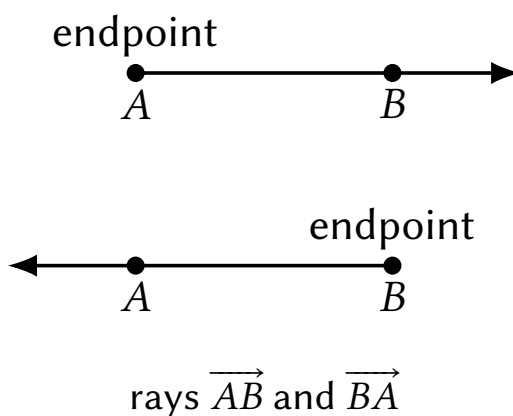
Segment: The line segment AB , or segment AB , (written as $\overline{AB}$) consists of the endpoints A and B and all points on $\overleftrightarrow{AB}$ that are between A and B .

Note that $\overline{AB}$ can also be named $\overline{BA}$.

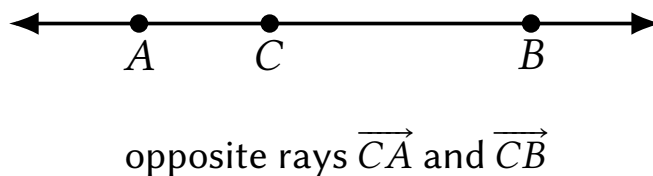


Ray: The ray AB (written as $\overrightarrow{AB}$) consists of the point A and all points on $\overleftrightarrow{AB}$ that lie on the same side of A as B .

Note that $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are different rays.



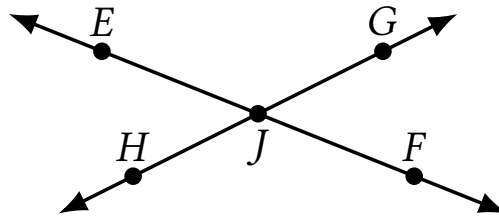
Opposite rays: If point C lies on $\overleftrightarrow{AB}$ between A and B , then $\overrightarrow{CA}$ and $\overrightarrow{CB}$ are opposite rays.



Segments and rays are collinear if they lie on the same line. So, opposite rays are collinear. Lines, segments, and rays are coplanar if they lie in the same plane.

Example 2

Name segments, rays, and opposite rays



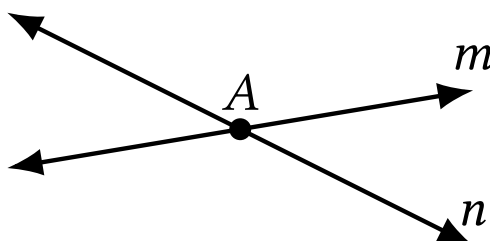
- (a) Give another name for $\overrightarrow{GH}$.
- (b) Name all rays with endpoint J . Which of these rays are opposite rays?
- (c) Give another name for $\overrightarrow{EF}$.
- (d) Are $\overrightarrow{HJ}$ and $\overrightarrow{JH}$ the same ray? Are $\overrightarrow{HJ}$ and $\overrightarrow{HG}$ the same ray?

Solution

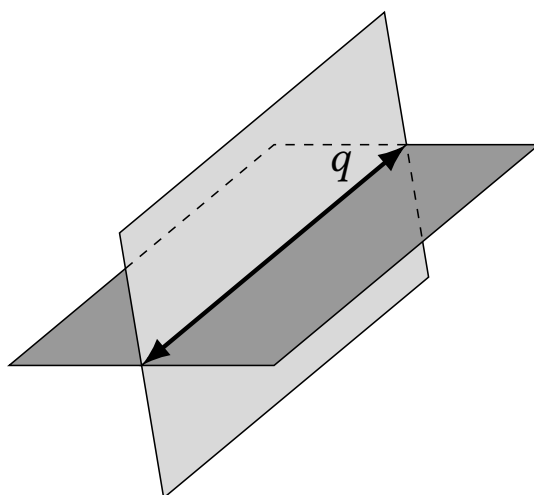
- (a) Another name for $\overrightarrow{GH}$ is $\overrightarrow{HG}$.
- (b) The rays with endpoint J are $\overrightarrow{JE}$, $\overrightarrow{JG}$, $\overrightarrow{JF}$, and $\overrightarrow{JH}$.
The pairs of opposite rays with endpoint J are $\overrightarrow{JE}$ and $\overrightarrow{JF}$, and $\overrightarrow{JG}$ and $\overrightarrow{JH}$.
- (c) Another name for $\overrightarrow{EF}$ is $\overrightarrow{FE}$.
- (d) $\overrightarrow{HJ}$ and $\overrightarrow{JH}$ are not the same rays. $\overrightarrow{HJ}$ includes the endpoint H and the points on $\overrightarrow{HJ}$ that lie on the same side as point J while $\overrightarrow{JH}$ includes the endpoint J and the points that lie on the same side as point H .
 $\overrightarrow{HJ}$ and $\overrightarrow{HG}$ are the same rays since they both contain the endpoint H and all the points of the line $\overrightarrow{HJ}$ that lie on the same side as point J .

Intersections

Intersection: Two or more geometric figures intersect if they have one or more points in common. The intersection of the figures is the set of points the figures have in common. Some examples of intersections are shown below.



The intersection of two different lines is a point.



The intersection of two different planes is a line.

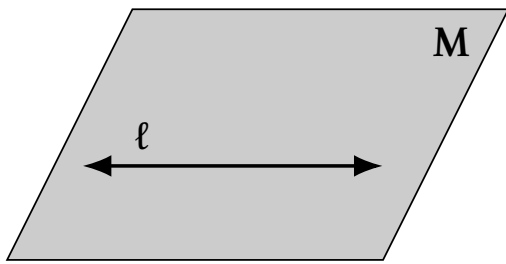
Example 3

Sketch intersections of lines and planes

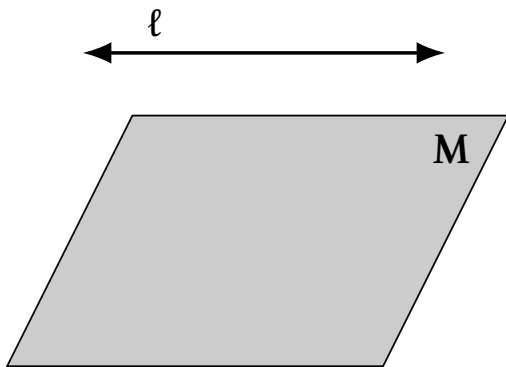
- (a) Sketch a plane and a line that is in the plane.
- (b) Sketch a plane and a line that does not intersect the plane.
- (c) Sketch a plane and a line that intersects the plane at a point.

Solution

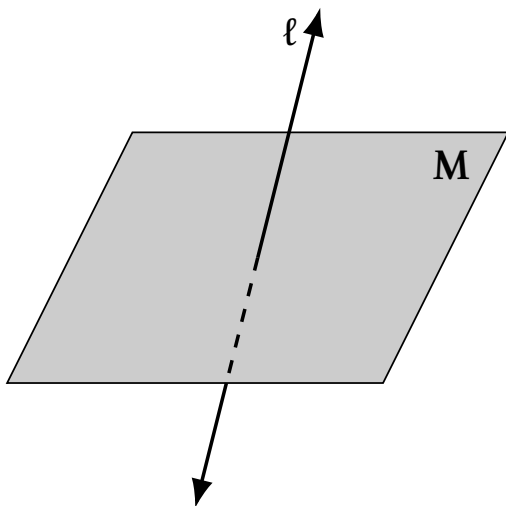
- (a) The line ℓ is in the plane M .



(b) The line ℓ does not intersect the plane M .



(c) The line ℓ intersects the plane M at a point.

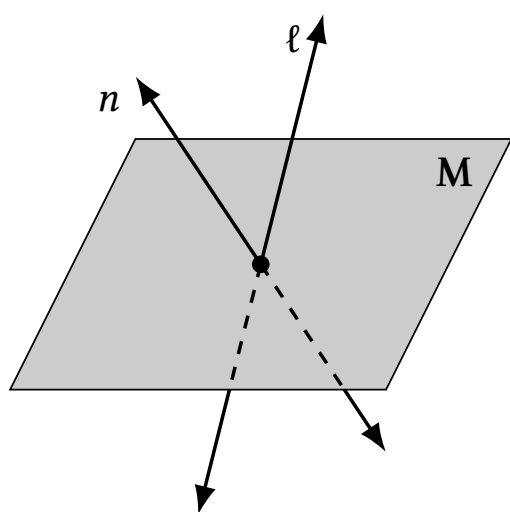


Example 4

Sketch two different lines that intersect a plane at the same point.

Solution

Lines l and n intersect the plane M at the same point.



Appendix

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